Advanced Databases - Lab No. 7

Third Year of the "Computer Engineering" Program

Oracle – Administration, Performance et transactions

Exercise 1: Tablespaces and Data Files - Creation, Quotas, and Management

Objective: Learn how to create and manage tablespaces, assign quotas, and understand how space limits affect users.

1. Connect as SYSTEM.
2. Create a new tablespace:

```
CREATE TABLESPACE tbs_students DATAFILE 'tbs_students1.dbf' SIZE 100M;
```

3. Create a new user with a quota:

```
CREATE USER student1 IDENTIFIED BY "123456"

DEFAULT TABLESPACE tbs_students

QUOTA 1M ON tbs_students;
```

4. Grant permissions:

```
GRANT CONNECT, RESOURCE TO student1;
```

5. Connect as student1 and create a table:

```
CREATE TABLE test_table (id NUMBER, name VARCHAR2(50));
```

6. Insert several rows until you reach the quota limit:

```
SET SERVEROUTPUT ON;

DECLARE
  v_counter NUMBER := 1;
BEGIN
  LOOP
    INSERT INTO test_table (id, name)
    VALUES (v_counter, 'Name ' || v_counter);  -- simple string
    v_counter := v_counter + 1;
    COMMIT;  -- commit to actually allocate space
```

```
  END LOOP;
EXCEPTION
  WHEN OTHERS THEN
    DBMS_OUTPUT.PUT_LINE('Tablespace quota exceeded after inserting ' ||
(v_counter-1) || ' rows.');
END;
/
```

7. Try to insert a new row and analyze the error message you get:

```
INSERT INTO test_table (id, name)
VALUES (99999, 'Hasna');
```

8. As SYSTEM, extend the quota to the user student1:

```
ALTER USER student1 QUOTA 5M ON tbs_students;
```

9. Try to insert the same previous row.

Exercise 2: Indexing and Optimization - B-Tree Index

Objective: Understand how a B-Tree index improves query performance and how to verify its use.

1. Connect as HR and create a copy of the EMPLOYEES table:

```
CREATE TABLE copy_employee AS SELECT * FROM employees;
```

2. Run the following query and record the execution time:

```
SELECT employee_id, first_name, last_name, salary, department_id

FROM copy_employee

WHERE department_id = 50

 AND salary > 5000

ORDER BY employee_id;
```

3. Create a B-Tree index:

```
CREATE INDEX btree_index ON copy_employee(department_id, salary, employee_id);
```

4. Run the same query again and record the new execution time:
5. Compare before and after results.
6. How does the B-Tree index affect query speed?

Exercise 3: Transactions and Concurrency Control - COMMIT, ROLLBACK

Objective: Learn how transactions work and how COMMIT and ROLLBACK affect data visibility.

1. Connect as HR and run:

SELECT * FROM employees WHERE department_id = 60;

2. Start a transaction:

UPDATE employees SET salary = salary + 500 WHERE department_id = 60;

3. Open a second session and run:

SELECT * FROM employees WHERE department_id = 60;

4. Observe if changes are visible.
5. Go back to session 1 and run:

COMMIT;

6. Check again in session 2, now the data should be visible.
7. Repeat another update, but this time use: ROLLBACK;
8. What does ROLLBACK do?